Mahalakshmi S

smmahasri@gmail.com | 6383731855 | Bengaluru

PROFESSIONAL SUMMARY

Software Engineer with 7+ years of hands-on expertise in System Design, Data Engineering, and developing scalable ETL pipelines. Proven track record delivering large-scale BI reporting solutions and regulatory data workflows in fast-paced, data-driven environments. Delivered measurable business impact through AI-driven development and infrastructure optimization.

SKILLS

- Programming & Scripting: Python, Shell Scripting, Unix
- Big Data & ETL: IBM DataStage, Data Warehousing, Spark
- Databases: MySQL, Teradata, DB2, Sybase
- Reporting & BI Tools: Tableau, Power BI
- Development Tools: IntelliJ, Visual Studio
- Orchestration & Scheduling: Airflow, AutoSys
- DevOps & Version Control: Jenkins, CI/CD pipelines, GIT, Bitbucket
- Tools & Utilities: Jira, Rally
- Cloud & Modern Data Stack (ELT) : Snowflake (Fundamentals), Azure Blob, DBT (Learning)

EXPERIENCE

Software Engineer- L3 Associate — Morgan Stanley *Jan 2022 – Present*

Bengaluru

Technologies: SQL, ETL, BI, Autosys, Python, Unix Scripting, Data Warehousing, Spark, GIT, Intellij, Visual Studio

Balance Sheet & Regulatory Reporting -APAC Region *Jan 2022 – Aug 2024*

- Designed, implemented, tested and maintained **ETL** components for **BI** reporting.
- Built and optimized stored procedures, database views, and **complex SQL** queries to improve data processing performance.
- Used **Apache Spark** for distributed data processing within ETL pipelines, enabling parallel transformation of large datasets.
- Developed regulatory and risk reporting solutions for Finance stakeholders, supporting business decision-making and regulatory compliance.
- Applied AI-assisted development such as **GitHub Copilot, Claude**, etc., across ETL pipelines and reporting workflows to increase developer productivity, optimize performance, and reduce implementation time.

Treasury CORE – Liquidity Risk & Regulatory Reporting -Global *Aug 2024 - current*

- Core team supporting liquidity risk, regulatory compliance, and treasury data management across the bank.
- Automated data workflows using **Airflow, Python and SQL(DB2)** improving operational efficiency.
- Automated approximately 90% of routine development tasks using **Claude** AI, significantly reducing manual coding effort and accelerating ETL development.
- Performed environment maintenance and data validations to support seamless UAT and regression cycles.
- Supported Agile delivery using Jira & Rally for sprint tracking, deployment and production readiness.
- Led and mentored a small team of engineers supporting the Treasury – Liquidity Risk & Regulatory Reporting platform, coordinating task allocation, conducting code reviews, providing technical guidance, and ensuring timely project delivery.

Infosys Limited – Senior Systems Engineer *Aug 2019 – Jan 2022*

Anti-Money Laundering - AML Team -Banking & Finance

Technologies: DataStage (ETL), MySQL DB, Teradata, Unix, AutoSys

- Optimized **IBM DataStage ETL** workflows by redesigning job stages, reducing unnecessary transformations, and tuning SQL queries, decreasing batch execution time from 2 hours to 40 minutes (67%).
- Created and scheduled **AutoSys** jobs and performed impact analysis on job flows.
- Worked on RAW and staging layers in **Data Warehousing** using **Teradata**.
- Performed unit/integration testing and supported **deployments**.

EDUCATION

B. Tech, Computer Science & Engineering — Sastra University *2015 – 2019*

CERTIFICATIONS

- Agile Development — LEX Infosys — 2021
- Hands-on training in Prompt Engineering, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI-assisted software development, and enterprise AI applications- AI For Business Leaders Bootcamp — OUTSKILL (INDBC)

INTERNSHIP

- Completed Internship at Infosys from 12/2018 to 04/2019, trained in Python, Linux, MySQL

PROJECT ACHIEVEMENTS

- **APAC Regulatory Reporting Automation**
  Partnered with APAC Australian Finance stakeholders to design and automate 20+ Balance Sheet regulatory reports for APRA (Australian Prudential Regulation Authority), reducing daily manual reporting effort by approximately 4 hours while improving reporting accuracy and compliance.
- **Data Validation & Audit Support**
  Designed and implemented enterprise-wide data validation and reconciliation processes across multiple regulatory reports, improving data quality, supporting audit readiness, and ensuring compliance with financial reporting standards.
- **AI-powered Cloud Migration**
  Leveraged Claude AI to accelerate the migration of nearly 1 TB of archival data files to Microsoft **Azure** blob, automating code generation, migration workflows, and validation activities while reducing manual effort and improving migration efficiency.

PERSONAL PROJECTS

- **Data Engineering & Semantic Search Pipeline | Python, Pandas, NumPy, SQL, Sentence Transformers**
  • Built a local Python data pipeline that turns messy technology-news articles into two outputs: a queryable company ARR observations over time, and a vector search index to support semantic search
  • Implemented semantic search using **Sentence Transformers (all-MiniLM-L6-v2)** with 384-dimensional embeddings and cosine similarity to identify related articles.
- **HR Assistant – Retrieval-Augmented Generation (RAG) Pipeline** | Python, RAG, Embeddings, LLM, Vector Search
  Built an **HR Assistant using a RAG pipeline**, leveraging document chunking, embeddings, vector search, and LLMs to provide context-aware answers from HR policy documents.